Quasi-frozen spin for both deuteron and proton beam at periodic EDM storage ring lattice

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A novel experimental setup with spatially separated magnetic and electrostatic fields is proposed for studying the electric dipole moment (EDM) of deuterons and protons. The design leverages a spin compensator to achieve precise measurements under ”quasi-frozen” spin conditions, demonstrating adaptability to existing facilities for research at higher energy range.

Keywords: Electric dipole moment, Quasi-frozen spin, Wien filter, Electrostatic deflector

I. INTRODUCTION

The search for the electric dipole moment (EDM) of fundamental particles, such as protons and deuterons, constitutes a critical endeavor in contemporary physics. The EDM offers a unique opportunity to probe CP violation and validate theoretical models that describe particle interactions beyond the framework of the Standard Model [1, 2]. For charged particles, the precise measurement of EDM necessitates the use of accelerators and the development of novel methodologies and technologies, requiring the adaptation of existing accelerator facilities to meet these experimental demands [3].

This paper proposes a versatile structure featuring by spatially separated magnetic and electrostatic fields, specifically designed to facilitate the study of EDM for both deuterons and protons. A notable feature of this structure is its ability to support experiments for both types of particles, thereby enhancing its applicability and efficiency. Special attention is given to utilization of Wien filters or electrostatic deflectors with kicker to compensate for the magnetic dipole moment (MDM) and establish conditions conducive to a quasi-frozen spin (QFS). The analysis demonstrates the adaptability of this structure for facility not originally designed for such experiments and explores its potential for advancing research at higher energy range and axion searches.

II. QUASI-FROZEN SPIN AT PERIODIC LATTICE

For an ensemble of particles, the spin can be described as a classical vector and its dynamics in external electromagnetic fields is described by Thomas-Bargmann-Michel-Telegdi (T-BMT) equation [4]

$$
\begin{aligned}
\frac{\mathrm{d}\vec{\mathrm{S}}}{\mathrm{d}t} &= \left(\vec{\Omega}_{\mathrm{MDM}} + \vec{\Omega}_{\mathrm{EDM}}\right) \times \vec{\mathrm{S}},\\
\vec{\Omega}_{\mathrm{MDM}} &= -\frac{q}{m\gamma}\Big\{(\gamma G+1)\vec{\mathrm{B}}_{\perp} + (G+1)\vec{\mathrm{B}}_{\parallel} -\\
&\quad - \left(\gamma G + \frac{\gamma}{\gamma+1}\right)\frac{\vec{\beta}\times\vec{\mathrm{E}}}{c}\Big\},\\
\vec{\Omega}_{\mathrm{EDM}} &= -\frac{q\eta}{2m}\left(\vec{\beta}\times\vec{\mathrm{B}} + \frac{\vec{\mathrm{E}}}{c} - \frac{\gamma}{\gamma+1}\frac{\vec{\beta}}{c}\left(\vec{\beta}\cdot\vec{E}\right)\right),
\end{aligned}
\tag{1}
$$

where $\vec{\Omega}_{\mathrm{MDM}}, \vec{\Omega}_{\mathrm{EDM}}$ – are the angular speed of rotation due to magnetic and electric dipole moment, $\vec{\mathrm{E}}, \vec{\mathrm{B}} = \vec{\mathrm{B}}_{\perp} + \vec{\mathrm{B}}_{\parallel}$ – external magnetic and electric field. m, q – mass and charge of a particle, γ is Lorenz factor. Dimensionless η factor is connected to the EDM value d and spin s of a particle: $d = \frac{\eta g}{2mc}s$, $G = \frac{g-2}{2}$ – anomalous magnetic moment.

For EDM measurements, it is first necessary to compensate for MDM-effect. The simplest approach, initially proposed by BNL [5], is based on the frozen spin method and uses an all-electric lattice, directly measuring spin precession due to a non-zero EDM in an radial electric field. This method requires the use of elements with pure electric field and the lattice must be specifically designed and operated for proton EDM research.

An alternative approach is the quasi-frozen spin concept, based on the principle of sequential spin compensation [6]. This leads to the spin oscillations relative to the momentum at one period and as a result at full ring.

A. Relative spin motion at QFS

Let us introduce the concept of QFS by examining orbital and spin rotation in a generalized framework. This approach is based on the dynamics governed by the Lorentz force for orbital motion and T-BMT equation for spin precession. It is essential to ensure first QFS condition that the orbital trajectory remains unperturbed, maintaining a closed orbit configuration.

$$
\Phi_p^{\mathrm{arc}} + \Phi_p^{\mathrm{comp}} = \frac{2\pi}{N},
\tag{2}
$$

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where index p states for momentum, Φ_p^{arc} and Φ_p^{comp} – total momentum rotation in arc and spin compensator element, N is the lattice periodicity. To facilitate a lattice suitable for both EDM studies and high-energy researches, it is essential to employ a pure magnetic arc with a guiding field, given the limitations of achievable electric field strengths. This way a corresponding spin compensator should be implemented and not perturb orbit at all. It can be formulated as $\Phi_p^{\mathrm{arc}} = \frac{2\pi}{N}$, then $\Phi_p^{\mathrm{comp}} = 0$. In order to compensate Lorenz force, the spin compensator must utilize both magnetic and electric fields. From the eq. 2 for spin compensation element with both magnetic and electric field

$$\Phi_{p\mathrm{B}}^{\mathrm{comp}} + \Phi_{p\mathrm{E}}^{\mathrm{comp}} = 0, \tag{3}$$

where $\Phi_{p\mathrm{B}}^{\mathrm{comp}}$ and $\Phi_{p\mathrm{E}}^{\mathrm{comp}}$ – total momentum rotation in magnetic and electric field at spin compensation. From T-BMT equation 1 it can be seen that maximum effectiveness on spin electric field then it is radial to the momentum. Moreover, the momentum rotation has also the maximum value for radial electric force, which means the most effectiveness for applied field

$$\vec{\Omega}_{p\mathrm{E}}^{\mathrm{max}} = \vec{\Omega}_{p\mathrm{E}_\perp} = \frac{q}{m\gamma}\frac{\vec{\beta}\times\vec{\mathrm{E}}_\perp}{c\beta^2}. \tag{4}$$

The second condition is to compensate MDM spin rotation in one period of the ring

$$\Phi_s^{\mathrm{arc}} + \Phi_s^{\mathrm{comp}} = 0, \tag{5}$$

where Φ_s^{arc} and Φ_s^{comp} – total spin rotation in arc and spin compensator. The ratio of spin rotation relative to orbital one as follows given by spin-tune in magnetic and electric field

$$\nu_{\mathrm{B}_\perp} = \gamma G, \quad \nu_{\mathrm{E}} = \gamma\beta^2\left(G - \frac{1}{\gamma^2-1}\right). \tag{6}$$

So, compensation occurs due to the fundamental difference between spin rotation in a magnetic and electric field. And the coupling equations for pure magnetic arc $\Phi_s^{\mathrm{arc}} = \nu_{\mathrm{B}_\perp}\Phi_p^{\mathrm{arc}}$ and magnetic $\Phi_{s\mathrm{B}}^{\mathrm{comp}} = \nu_{\mathrm{B}_\perp}\Phi_{p\mathrm{B}}^{\mathrm{comp}}$ and electrical $\Phi_{s\mathrm{E}}^{\mathrm{comp}} = \nu_{\mathrm{E}}\Phi_{p\mathrm{E}}^{\mathrm{comp}}$ in spin compensator. Than for eq. 5

$$\begin{aligned}\nu_{\mathrm{B}_\perp}\Phi_p^{\mathrm{arc}} + \left(\nu_{\mathrm{B}_\perp}\Phi_{p\mathrm{B}}^{\mathrm{comp}} + \nu_{\mathrm{E}}\Phi_{p\mathrm{E}}^{\mathrm{comp}}\right) &= \\ = \nu_{\mathrm{B}_\perp}\left(\Phi_p^{\mathrm{arc}} + \Phi_{p\mathrm{B}}^{\mathrm{comp}}\right) + \nu_{\mathrm{E}}\Phi_{p\mathrm{E}}^{\mathrm{comp}} &= 0.\end{aligned} \tag{7}$$

Finally, from first and second condition can be received for rotation via magnetic or electric field in spin compensator element

$$\Phi_{p\mathrm{E}}^{\mathrm{comp}} = -\Phi_{p\mathrm{B}}^{\mathrm{comp}} = \Phi_p^{\mathrm{arc}}\frac{\gamma^2 G}{G+1} = \frac{2\pi}{N}\frac{\gamma^2 G}{G+1}. \tag{8}$$

No mention was made of the physical structure of the compensation element, only of the integral features of the presented field components.

B. Spin compensation element length

Spin compensation element length can be calculated for magnetic and electric field

$$L = \Phi_{p\mathrm{B}}^{\mathrm{comp}} R_{\mathrm{B}} = \Phi_{p\mathrm{E}}^{\mathrm{comp}} R_{\mathrm{E}}, \tag{9}$$

where $R_{\mathrm{B}}, R_{\mathrm{E}}$ – radius of magnetic and electric field. In general, total radius for element with combined magnetic and electric fields can be find as

$$\frac{1}{R} = \frac{1}{R_{\mathrm{B}}} + \frac{1}{R_{\mathrm{E}}}, \quad R_{\mathrm{B}} = \frac{\mathrm{B}\rho}{\mathrm{B}}, \quad R_{\mathrm{E}} = \frac{\kappa}{\mathrm{E}}, \tag{10}$$

where $\mathrm{B}\rho = \dfrac{p_0}{e}$ – magnetic rigidity, $p_0 = \gamma m\beta c$, $\kappa = \dfrac{p_0\beta c}{e}$ – electrical rigidity. If we consider no orbital distortion in total, Lorentz force should be zero, so $R = 0$ and $R_{\mathrm{B}} = -R_{\mathrm{E}}$.

The limitation on the strength of the electric field is much more significant than in the case of a magnetic field. According to the eqs. 8 and 9 can be received the length of compensator element per one period

$$\begin{aligned}L_{\mathrm{min}} = \Phi_{p\mathrm{E}}^{\mathrm{comp}} R_{\mathrm{E}} = \frac{2\pi}{N}\frac{\gamma^2 G}{G+1}\frac{\kappa}{\mathrm{E}_{\mathrm{max}}} &= \\ = \frac{2\pi}{N}\frac{G}{G+1}\frac{mc^2}{e}\frac{\gamma(\gamma^2-1)}{\mathrm{E}_{\mathrm{max}}}&,\end{aligned} \tag{11}$$

where $\mathrm{E}_{\mathrm{max}}$ – maximum of electric field, that limit the minimal achievable length regardless the researched particle. For total length $L = L_{\mathrm{min}}^{\mathrm{total}} = N \cdot L_{\mathrm{min}}$ and is independent from number of periods. Maximum attainable electric fields are 10-13 V/m [7], this corresponds to a magnetic field of the order of 70-90 mT accordingly, it will be used later for evaluation.

III. FEATURES OF PROTON AND DEUTERON

A. Dependence on beam parameters

To effectively analyze spin dynamics, it is essential to maximize the analyzing power of the polarimeter. Thus, the beam energy plays a critical role in achieving sufficient experimental statistics for this purpose. The maximum figure of merit of the polarimeter is reached at a proton beam energy of 270 MeV [8] and the same is true for deuterons [9].

Additionally, orbital and spin dynamics vary among different types of particles. First, the mass-to-charge ratio of deuterons is twice that of protons. Second, the anomalous magnetic moments for protons and deuterons are notably different: $G_{\mathrm{p}} = 1.79$ for protons and $G_{\mathrm{d}} = -0.14$ for deuterons. These differences are shown at Table 1 and are necessitate tailored approaches for each particle type when designing experiments and analyzing results.

In a magnetic field, according to eq.6 the direction of rotation depends solely on the sign of the anomalous magnetic

TABLE 1. Particles and experiment parameters.

Particle	A/Z	G	γ_{exp}	E_{exp}, MeV/u	L, m
Deuteron d	2	-0.14	1.144	135	53
Proton p	1	1.79	1.289	270	247
			1.078	73	53

moment and is independent of energy. However, the magnitude of the relative rotation is influenced by both the energy and absolute value of anomalous magnetic moment. For protons this effect is so important that a care must be taken to assure the spin rotation per arc $\gamma G \cdot \Phi_p^{\text{arc}} < \pi/2$ to preserve accumulation of the EDM effect for longitudinally polarized beam, so that QFS lattices only with $N = 8$ or larger give a chance to measure the proton EDM.

More precisely, the EDM measurement slightly deviates from the frozen spin condition, in the first order decreases as given by the factor [10]

$$J_0(\Phi_s^{\text{arc}}) = 1 - \frac{\Phi_s^{\text{arc}\,2}}{4}, \tag{12}$$

where Φ_s^{arc} represents angle associated with the MDM-effect at arc in period. For various particle and structure periodicity it is cited in Table 2. The maximum periodicity considered is $N = 16$, due to lattice design limitations. Shown that for stored deuteron lattices the quasi-frozen spin and frozen-spin are close to each other. For protons, 16-periodic structure offers a real opportunity of practical tests of the QFS approach.

TABLE 2. Parameters of lattices and experiment at beam energy of 270 MeV.

Particle	N	Φ_s^{arc}, deg	$J_0(\Phi_s^{\text{arc}})$
Deuteron d	2	-29.45	0.934
	4	-14.72	0.983
	8	-7.36	0.996
	16	-3.68	0.999
Proton p	8	103.83	0.179
	16	51.91	0.795

In electrostatic fields, the behavior of protons and deuterons is further complicated by their distinct interactions with the field. For protons, the direction of spin rotation can depend on the energy, particularly around the "magic energy" where $\gamma_{\text{mag}} = \sqrt{\dfrac{G_{\text{p}}+1}{G_{\text{p}}}} = 1.248$ ($E_{\text{mag}} = 233$ MeV). At any way the influence on proton is much lower than for deuterons as shown at Fig. 1. For deuterons, however, there is no equivalent "magic energy," and their spin rotation behavior remains consistent across energy range.

These fundamental differences lead to several important consequences for lattice construction. Specifically, modifications are necessary to accommodate the study of both particle types. According to eq. 8 the sign of rotation for proton and deuteron differs. This shows the necessity of either switching the polarity of the spin compensator's fields or rotating

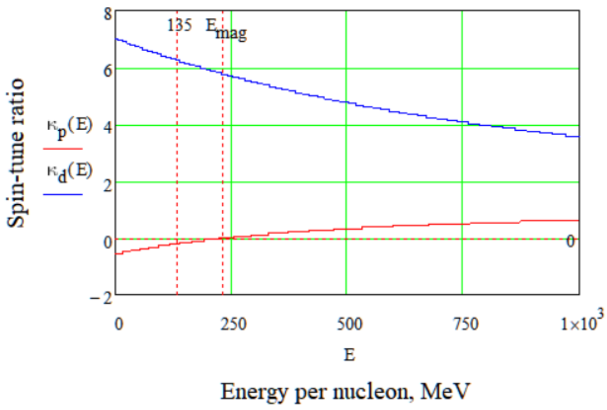

Fig. 1. Spin-tune ratio $\kappa = \dfrac{\nu_{\text{B}_\perp}}{\nu_{\text{E}}}$ for deuteron and proton.

it by π along the longitudinal axis. Additionally, the minimal achievable spin compensator length as given by eq. 11, is longer for proton than for deuterons at optimal polarimeter energy. This discrepancy can be addressed by lowering the experimental energy for protons (Fig. 2). At a spin compensator length equivalent to that used for deuterons, the proton energy should be reduced to 73 MeV, the analyzing power decreases by a factor of 2-3. Two viable approaches for arranging additional EM-fields are the implementation of a Wien filter or an electrostatic deflector with a kicker.

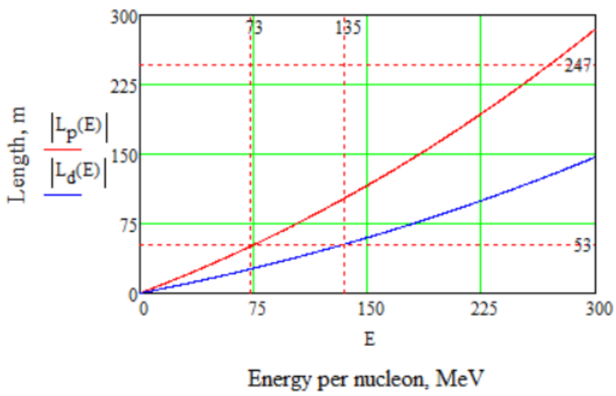

Fig. 2. Total length of spin compensator vs Energy per nucleon in order to make QFS lattice for deuteron and proton.

B. Wien filter

The Wien filter is a crucial tool in a particle accelerator, designed to manipulate spin dynamics without altering the orbital trajectory of particles by utilizing perpendicular electric and magnetic fields that cancel each other out, resulting in zero net Lorentz force. As EDM-term is proportional to Lorenz force, Wien filter not violate spin-vector. This configuration allows particles to move in a straight line while enabling precise control over their spin orientation, making it ideal for experiments requiring spin manipulation. The filter's design, ensures no net deflection along the beamline, minimizing additional space requirements in the ring's straight sections and allowing for a classical sequential arrangement of ring elements.

Adaptability is evident in its use for both deuterons and protons; for deuterons, it can be employed directly at optimal energies, while for protons, the polarity must be switched or the setup rotated by π along the longitudinal axis and used at lower energies (Fig. 3). This flexibility facilitates the development of methodologies for studying spin dynamics across different particle types, enhancing experimental strategies. Overall, the Wien filter's ability to control particle spin without disrupting orbital paths makes it an invaluable component in advanced accelerator experiments.

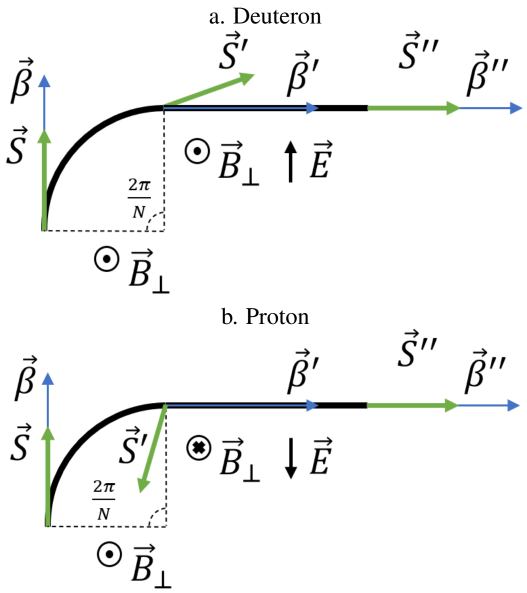

Fig. 3. One period conceptual scheme of QFS lattice with Wien filters for a. deuterons, b. protons.

C. Electrostatic deflector with kicker

Pure electrostatic deflector designed to manipulate particle trajectories and spin dynamics by creating a radial electric field with non-zero curvature. Unlike the Wien filter, this deflector requires an additional kicker to compensate for orbital deviations, which increases the overall length of the compensator. This setup is particularly useful for creating bypass sections in addition to native straight sections, allowing for flexible experimental configurations (Fig. 4). But it should be considered distance between the spatial assignment of equipment to enable option with independent section. However, it necessitates careful consideration of the spatial arrangement of equipment to enable independent section operation. A key challenge is that each type of particle requires a different curvature, which must be determined at the outset of lattice design. Despite these complexities, the electrostatic deflector with kicker offers a robust solution for controlling particle spin and trajectory.

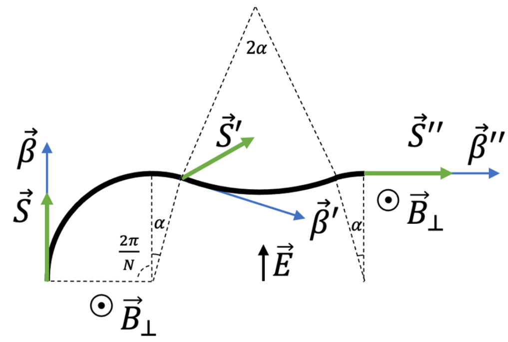

Fig. 4. One period conceptual scheme of QFS lattice with electrostatic deflector and kicker for deuterons.

IV. FURTHER RESEARCHES

Currently, the proposed energy requirements for EDM studies are set at 270 MeV and lower, but specific conclusions regarding arc rotation have yet to be drawn. The primary condition is to achieve a momentum rotation at a specific angle Φ_p^{arc} with a corresponding Φ_s^{arc} for spin. This can be accomplished by carefully selecting the magnetic field integral to ensure precise alignment of momentum and spin dynamics. To maximize effectiveness, the magnetic field should be as strong as possible to minimize the effective arc length, considering the unique features of the lattice design. This approach not only optimizes studies at current energy range but also opens the possibility for other polarized or non-polarized beam research at higher energies, potentially reaching several GeV. The versatility of the QFS structure allows it to be implemented in various lattice designs, including existing accelerator rings, thereby facilitating advanced studies at elevated energy range.

Wien filters do not affect beam dynamics and influence only the spin motion that helps to convert spins in storage ring into broadband axion antenna [11, 12].

V. SUMMARY

This article presents the implementation of a quasi-frozen spin structure for a periodic lattice to study the Electric Dipole Moment of charged light particles. The design achieves precise spin compensation of MDM-effect per one period by employing elements with matched curvatures for electric and magnetic fields, allowing for effective spin manipulation. Wien filters, which do not cause beam deflection, and electrostatic deflectors, which enable bypass creation, are integral to this flexibility. An expression for the optimal spin compensator length is derived. Notably, the setup allows for the investigation of proton EDM using the same ring as for deuterons, simply by switching the polarity or rotating the spin compensator by π and operating at reduced energy. The study demonstrates the versatility of the quasi-frozen spin lattice, which can be adapted for multiple purposes, including higher-energy researches within existing facilities and axion searches.

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